

Mean, Variance, & Covariance Handout

If integration is required for the problem, show the integration.

1. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed 3 times. *Use the probability distribution table created as the answer to #1 in Mod3 Probability Distributions Handout.*
 - a. Find the mean of X .
 - b. Find the variance of X .
 - c. Find the standard deviation of X .
2. The local animal shelter adopts cats and dogs each week with the following probabilities.

X	3	4	5	6	7	8
P(x)	0.15	0.3	0.25	0.18	0.1	0.02

- a. Find the mean number of cats and dogs adopted each week.
 - b. Find the variance for the number of cats and dogs adopted each week.
 - c. Find the standard deviation for the number of cats and dogs adopted each week.
3. In a gambling game a person draws a single card from a standard deck of cards. A person is paid \$15 for drawing a jack or a queen and \$5 for drawing a king or an ace. A person who draws any other card type pays \$4. If a person plays this game, what is the expected gain?
4. The probability distribution function of X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

If $E(x) = 3/5$, find a and b .

5. A random variable X , which represents the weight (in ounces) of an article, has the probability distribution function,

$$f(x) = \begin{cases} x - 8, & 8 \leq x \leq 9 \\ 10 - x, & 9 < x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

- a. Calculate the mean of the random variable X .
 - b. Calculate the variance of the random variable X .

6. A product is classified according to the number of defects it contains and the factory that produces it. Let X and Y be the random variables that represent the number of defects per unit and the factory number, respectively. The entries in the table represent the joint probability distribution function of a randomly chosen product. *Use the marginal distributions created for #1 in Mod4 Bivariate Probability Distributions Handout.*

		Y	
		1	2
X	0	1/8	1/16
	1	1/16	1/16
	2	3/16	1/8
	3	1/8	1/4

- Find the expected value of X .
 - Find the expected value of Y .
 - Find the variance of X .
 - Find the variance of Y .
 - Find the covariance of X and Y .
 - Find the correlation between X and Y .
7. The joint probability distribution function of X and Y is given by the following. *Use the marginal distribution created in the answer to #2 in Mod4 Bivariate Probability Distributions Handout.*

$$f(x) = \begin{cases} \frac{6}{7} \left(x^2 + \frac{xy}{2} \right), & 0 < x < 1, \quad 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- Find the expected value of X .
- Find the expected value of Y^2 .
- Find the expected value of X^2Y .